

Which City is Rainier? Seattle, WA or Portland, OR?

Introduction

Someone who wants to move to a new city might want to know more about the weather for their potential options. An important aspect of the weather for daily life is how rainy a city might be.

The question of rain is somewhat vague and might cover multiple different measures of rain. For this analysis, we will explore the difference in average rainfall, as well as the average number of rainy days.

The data used

We will be using data gathered from a single weather station in each city across the date range of January 1st, 2018 to December 31st, 2022.

The Seattle data was accessed through the GitHub repository for DATA 5100.

The Portland data was accessed through a request made through NOAA's Climate Data Online Search tool.

Data analysis methods

We will measure average rainfall as mean precipitation in inches and average number of rainy days as the proportion of days with measured precipitation.

Missing values were imputed using the average value for that measure for that day of the year across each other available year.

In order to get a clearer understanding of how seasons may affect weather patterns, mean precipitation and proportion of rainy days were calculated for each month across all years.

A t-test was performed between the mean precipitation values for each corresponding month for each city, and a z-test was performed between the proportional value of days with measured precipitation for each corresponding month for each city to determine any statistically significant differences in either measure

Results

Average rainfall—measured as the mean precipitation in inches—for each month of the year for each city was calculated and plotted as a bar chart (figure 1). The only months with statistically significant differences are July and August, marked with a ‘*’, and show that Portland received significantly fewer inches of precipitation in those months.

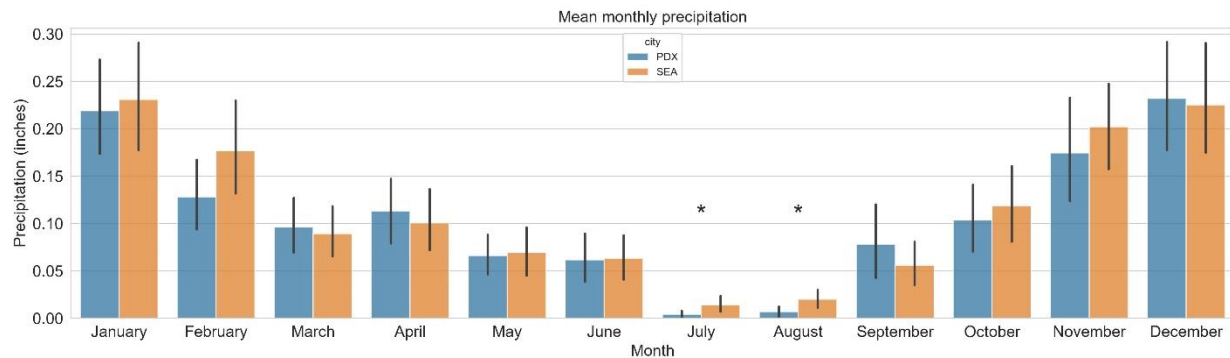


Figure 1. A bar plot of the mean monthly precipitation shows significant differences between Seattle and Portland only for July and August. Error bars display a 95% confidence interval.

The proportion of days each month that had rain—understood as a non-zero precipitation value for that day—for each city was calculated and plotted as a bar chart (figure 2). Statistically significant differences were found for eight months out of the year, each marked with a ‘*’, and show that Portland had significantly fewer rainy days in those months.

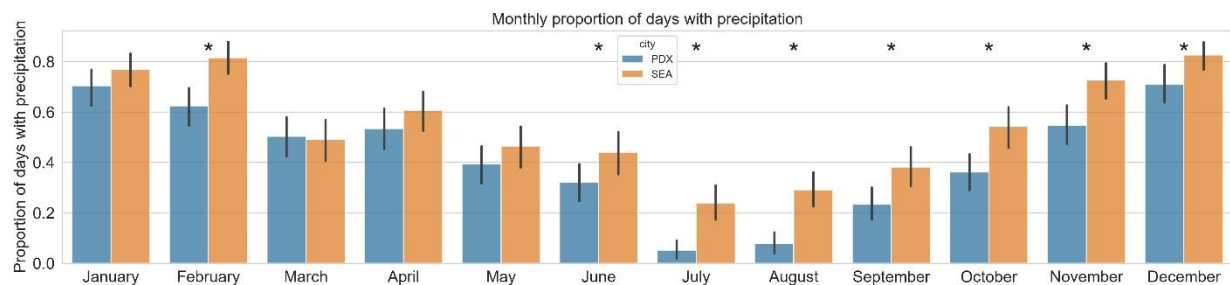


Figure 2. A bar plot of the proportion of days each month with measured precipitation shows significant differences between Seattle and Portland for 8 months out of the year. Error bars display a 95% confidence interval.

Conclusion

Given that the mean monthly precipitation was broadly comparable between SEA and PDX, but Portland had significantly fewer days with measurable precipitation, we can infer that Portland must receive more inches of rain on the days that it does rain.

Perhaps the best answer to our original question—which city is rainier—is to say that Portland has fewer rainy days, but that those rainy days are rainier.